

TPC-408: Complete-Shell C1 Q-Scale Extension

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Abstract

TPC-407 tested a locally normalized adjacent-entry proxy on four even complete prime shells. This note extends the finite audit to the next two scales, $Q = 65536$ and 131072 , whose complete shells contain respectively 5709 and 10749 primes. The odd cardinalities are retained in full by using the explicit alternating index profile with $m_- = \lfloor r/2 \rfloor$ and $m_+ = \lceil r/2 \rceil$. The exact local identity and Cauchy–Schwarz proof still give $0 \leq z \leq t_1/(a_{\min}\sqrt{S_0S_1}) \leq 4/(a_{\min}H) \leq 4/H$. An exact rational certificate and independent literal CRT masked-energy replay verify both rows. This remains one synthetic proxy entry and makes no arithmetic or twin-prime claim.

1 Odd complete-shell extension

Fix $H = 66$ and $N = 264 = 4H$. For $Q > N$, let the complete shell $(Q, 2Q]$ contain $r \geq 2$ primes $p_0 < \cdots < p_{r-1}$. We retain all of them, including odd r , and impose

$$o \equiv 0 \pmod{p_i} \quad (i \text{ even}), \quad o \equiv -N \pmod{p_i} \quad (i \text{ odd}).$$

Choose the CRT representative above 10^6 . Put

$$t_d = \frac{H^2}{H^2 + d^2}, \quad S_0 = \sum_{d=1}^{N-1} t_d^2, \quad S_1 = \sum_{d=1}^{N-2} t_d^2 + t_1^2,$$

and $a_i = p_i^3/[Q^2(p_i - 1)]$. If $m_- = \lfloor r/2 \rfloor$ and $m_+ = \lceil r/2 \rceil$, let P_- be the odd-index amplitude sum and V_-, V_+ the odd/even square sums. The TPC-404 local proxy gives

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

The height H here is a proxy parameter, not the physical h_0 .

2 Finite theorem

Theorem. With $a_{\min} = \min_i a_i$ and $z = M/\sqrt{G_0G_1}$,

$$0 \leq z \leq \frac{t_1}{a_{\min}\sqrt{S_0S_1}} \leq \frac{4}{a_{\min}H} \leq \frac{4}{H}. \quad (1)$$

Proof. Since $p_i > Q > N$, an even prime masks offset zero and an odd prime first masks offset N ; no shell prime masks offset one. Thus the three identities above hold whether r is even or odd.

In particular $G_1 \geq V_- S_1$. Cauchy–Schwarz and the odd-index lower bound give $P_-^2 \leq m_- V_-$ and $V_- \geq m_- a_{\min}^2$. Therefore

$$z^2 \leq \frac{t_1^2 m_-}{V_- S_0 S_1} \leq \frac{t_1^2}{a_{\min}^2 S_0 S_1}.$$

For $1 \leq d \leq H$, $t_d \geq 1/2$, and those terms occur in both sums, so $S_0, S_1 \geq H/4$. Also $a_i = (p_i/Q)^2 p_i / (p_i - 1) > 1$. Taking square roots proves (1). $\square$

3 Exact certificate

The producer enumerates every prime in each declared shell and performs integer CRT and rational arithmetic. The two shell counts are

Q	r	m_-	m_+	normalized entry
65536	5709	2854	2855	0.005213967369619
131072	10749	5374	5375	0.005218048074517

The corresponding H observations are 0.344121846394880 and 0.344391172918121. These decimals are finite float64 observations only; the certificate stores exact rational squares, CRT periods, and all shell primes. The independent checker reconstructs the sieve and literally replays every per-prime, per-coordinate mask before comparing both row energies and the adjacent coefficient.

4 Route boundary

This is a finite extension of one synthetic adjacent normalized proxy entry. It is not a full normalized operator estimate, a physical h_0 theorem, an arithmetic sign or L^2 theorem, a fixed-power saving, Route-A or Route-B closure, or a twin-prime result. The claim status is the exact finite complete-shell Q-scale extension recorded in the certificate; arithmetic advance is **NO**, fixed-power credit is 0, and full Gate B is **OPEN**.

Reproduction

Run the producer, independent replay, and adversarial stress checker with `--check` in both normal and optimized Python modes. Bridge-B repeats these checks, requires empty stderr and identical output, and locks every release artifact by SHA-256.